JupySQL

Biblio

ipython-sql

- https://pypi.org/project/ipython-sql/
- https://www.geeksforgeeks.org/how-to-install-ipython-sql-package-in-jupyter-notebook/

JupySQL

- Integración MariaDB
- Integración MySQL
- Integración DuckDB
- Integración Clickhouse

PyMySQL: Pure Python MySQL/MariaDB Driver

- https://pypi.org/project/PyMySQL/
- https://pymysql.readthedocs.io/en/latest/index.html
- https://www.geeksforgeeks.org/how-to-install-pymysql-in-python/

MySQL Connector/Python

- https://pypi.org/project/mysql-connector-python/
- https://dev.mysql.com/doc/connector-python/en/connector-python-installation.html

MariaDB Connector/Python

- https://pypi.org/project/mariadb/
- https://mariadb.com/resources/blog/how-to-connect-python-programs-to-mariadb/

Demo: *Sakila DB* en *MariaDB/MySQL* con *Jupyter*

```
%load_ext sql
%config SqlMagic.autopandas=True
```

```
import pymysql
import pandas as pd
```

```
%sql?
```

MariaDB JupySQL

```
%sql mysql+pymysql://jupyter:H4!b5at22@mariadb/sakila --alias mariadb
#%sql mariadb
%sql --connections
```

Active connections:

current	url	alias
*	mysql+pymysql://jupyter:***@mariadb/sakila	mariadb

```
%sqlcmd tables
```

Name
actor
address
category
city
country
customer
film
film_actor

Name
film_category
film_text
inventory
language
payment
rental
staff
store

```
%%sql
SELECT * FROM actor
```

	actor_id	first_name	last_name	last_update
0	1	PENELOPE	GUINESS	2006-02-15 04:34:33
1	2	NICK	WAHLBERG	2006-02-15 04:34:33
2	3	ED	CHASE	2006-02-15 04:34:33
3	4	JENNIFER	DAVIS	2006-02-15 04:34:33
4	5	JOHNNY	LOLLOBRIGIDA	2006-02-15 04:34:33
...	...	...	...	...
198	199	JULIA	FAWCETT	2006-02-15 04:34:33
199	200	THORA	TEMPLE	2006-02-15 04:34:33
200	201	ADW	ADE	2006-02-15 01:31:33
201	202	ADW	GII	2006-02-15 02:31:33
202	203	IBD	GCD	2006-02-15 03:31:33

```
%sql --close mariadb
```

PyMySQL

```python
conn = pymysql.connect(
    host='mariadb',
    port=3306,
    user='jupyter',
    password='H4!b5at22',
    charset='utf8mb4',
    cursorclass=pymysql.cursors.DictCursor
)
```

```python
# Cursor query
cur = conn.cursor()
cur.execute("select title, description, release_year from sakila.film limit 20")
output = cur.fetchall()

for i in output:
    print(i)

# To close the connection
conn.close()
```

MariaDB Connector/Python

```python
# Module Import
import mariadb
import sys

# Print List of Customers
def print_customers(cur):
     """Retrieves the list of actors from the database and prints to stdout"""

     # Initialize Variables
     customers = []

     # Retrieve customers
     cur.execute("SELECT first_name, last_name, email FROM sakila.customer LIMIT 20")

     # Prepare customers
     for (first_name, last_name, email) in cur:
        customers.append(f"{first_name} {last_name}: <{email}>")

     # List customers
     print("\n".join(customers))

# Instantiate Connection
try:
     conn = mariadb.connect(
        host="mariadb",
        port=3306,
        user="jupyter",
        password="H4!b5at22")

     # Instantiate Cursor
```

```
     cur = conn.cursor()

     print_customers(cur)

     # Close Connection
     conn.close()

except mariadb.Error as e:
      print(f"Error connecting to the database: {e}")
      sys.exit(1)
```